

HNSS07 · The Brain — Anatomy & Function Chart

Anatomy and functions of the telencephalon, diencephalon, midbrain, pons & medulla, and cerebellum, as presented in lecture | **STEP 1** = beyond class content · **EXAM EMPHASIS** = flagged in class as testable

HoyaDocs
Updated Oct 1, 2026 · Rev 1

Brain Divisions at a Glance · Telencephalon · Diencephalon

Telencephalon
cortex + subcortical nuclei + white matter

Diencephalon
thalamus · hypothalamus · subthalamus · epithalamus

Midbrain
tectum + tegmentum (upper brainstem)

Pons & medulla
lower brainstem; CN V–XI nuclei

Cerebellum
2 hemispheres, vermis, 4 deep nuclei

Context: development, orientation, sections, meninges

TOPIC	KEY FACTS
Development	Prosencephalon → telencephalon + diencephalon · Mesencephalon → midbrain · Rhombencephalon → metencephalon (pons + cerebellum) + myelencephalon (medulla)
Orientation	Forebrain: superior–inferior = dorsal–ventral ; anterior–posterior = rostral–caudal . In the brainstem, dorsal–ventral runs front-to-back
Sections	Coronal, sagittal, horizontal (axial) — use the corpus callosum as your landmark
Meninges (out → in)	Dura (two thick layers; 4 infoldings such as the falx cerebri; contains venous sinuses) → arachnoid (cushions; subarachnoid space holds CSF) → pia (adheres to the brain)

Telencephalon (cerebrum)

STRUCTURE	ANATOMY (WHAT'S IN IT)	FUNCTION	CLINICAL LINK (LECTURE)
Cerebral cortex <i>the thinking surface</i> GRAY MATTER	Two hemispheres , each with four lobes : frontal, parietal, temporal, occipital (named like the overlying skull bones). Lobes contain substructures, e.g., the hippocampus (temporal lobe)	Cognitive + affective (language, memory, thought, emotion); sensory (vision, audition, taste, smell, touch, pain); motor . Primary areas: vision = occipital, hearing = temporal, touch = parietal	Broca's aphasia (Broca's area): speech + language, memory, executive, motor deficits
Subcortical nuclei <i>the deep gray matter</i> GRAY MATTER	Amygdala ; basal ganglia (caudate and others)	Support cortical functions; basal ganglia = motor and cognitive (memory, executive, language)	Huntington (caudate): chorea + cognitive deficits. Basal ganglia are NOT just motor EXAM EMPHASIS
White matter <i>the highways</i> TRACTS	Tracts (e.g., internal capsule) and commissures (e.g., corpus callosum , joining the hemispheres)	Connect gray matter structures throughout the brain and nervous system; seen on diffusion tensor imaging (shows tracts, not direction)	White matter damage can disrupt a whole network

Diencephalon (one in each hemisphere)

STRUCTURE	ANATOMY (WHAT'S IN IT)	FUNCTION	CLINICAL LINK (LECTURE)
Thalamus <i>the relay and gatekeeper</i> DIENCEPHALON	Many nuclei (e.g., lateral geniculate nucleus for vision STEP 1)	Relay and gatekeeper with the cortex: sensory, motor, cognitive, limbic	—
Hypothalamus <i>the 4 F's + sleep</i> DIENCEPHALON	Various substructures, including the pituitary	Feeding, Feelings, Fever, Mating (the “four F’s”) + Sleep	Pituitary adenoma (17%): bitemporal hemianopsia (optic chiasm) + Cushing (↑ ACTH)
Subthalamus <i>the basal ganglia partner</i> DIENCEPHALON	Subthalamic nucleus	Functionally part of the basal ganglia : motor, memory, language	Developmentally diencephalon, functionally basal ganglia
Epithalamus <i>the clock</i> DIENCEPHALON	Habenula + pineal gland	Habenular nuclei: diverse behavioral functions. Pineal : circadian rhythms (melatonin)	—

★ **USMLE high-yield**: Telencephalon = **two hemispheres, four lobes** (hippocampus in the temporal lobe), subcortical **amygdala + basal ganglia**, and white matter (**internal capsule, corpus callosum**). Functions: cognition/affect, senses, movement. Basal ganglia are **not just motor** **EXAM EMPHASIS**. Most functions rely on **networks**, not one structure. **Thalamus** = relay + gatekeeper; **hypothalamus** = feeding, feelings, fever, mating + sleep (includes pituitary); **subthalamic nucleus** = functionally basal ganglia; **pineal** = circadian rhythm.

Midbrain · Pons & Medulla · Cerebellum

Midbrain (upper brainstem)

STRUCTURE	ANATOMY (WHAT'S IN IT)	FUNCTION	CLINICAL LINK (LECTURE)
Tectum <i>the roof</i> (MIDBRAIN (DORSAL))	Inferior colliculus, superior colliculus	Inferior = audition ; superior = vision (plus other roles)	Inferior = ears (lears); Superior = sight
Reticular formation <i>the arousal network</i> (TEGMENTUM)	Midbrain reticular formation	Motor and cardiovascular control, pain modulation , sleep & consciousness	—
Red nucleus <i>the gait coordinator</i> (TEGMENTUM)	Tegmental nucleus	Motor coordination, gait	—
Periaqueductal gray (PAG) <i>the pain dimmer</i> (TEGMENTUM)	Around the cerebral aqueduct	Autonomic function, pain modulation	—
Raphe nuclei <i>the thermostat</i> (TEGMENTUM)	Midline nuclei	Thermoregulation (serotonin source STEP 1)	—
Ventral tegmental area <i>the reward center</i> (TEGMENTUM)	VTA	Reward-based learning (dopamine STEP 1)	—
Substantia nigra <i>the dark nucleus</i> (TEGMENTUM)	Pars compacta (dopaminergic); connected with the basal ganglia	Motor, memory, language — works with the rest of the basal ganglia	Parkinson disease : SN pars compacta dopamine neurons degenerate

Midbrain = upper brainstem. Other tegmental nuclei named in lecture: Edinger-Westphal, pedunculo pontine, interpeduncular, parabrachial pigmented. Cranial nerve nuclei III–IV sit here **STEP 1**.

Pons & medulla (lower brainstem)

STRUCTURE	ANATOMY (WHAT'S IN IT)	FUNCTION	CLINICAL LINK (LECTURE)
Pons <i>the bridge</i> (LOWER BRAINSTEM)	Cranial nerve nuclei V–VIII (trigeminal, abducens, facial, vestibulocochlear); locus coeruleus = main brain source of norepinephrine	Breathing, equilibrium, hearing, taste, facial touch + pain, eye movement, facial expression, chewing, swallowing, saliva + tears	Pontine stroke : CN signs, double vision, balance problems, weakness on both sides
Medulla (oblongata) <i>the life support</i> (LOWER BRAINSTEM)	Cranial nerve nuclei, e.g., IX, X, XI ; arcuate nucleus (controls breathing rate); joins the spinal cord	Autonomic : breathing, heart rate, blood pressure, swallowing, sleep cycle, vomiting, sneezing, coughing; relays brain ↔ spinal cord	You cannot survive without a medulla — it controls the involuntary (autonomic) functions EXAM EMPHASIS

Cerebellum

STRUCTURE	ANATOMY (WHAT'S IN IT)	FUNCTION	CLINICAL LINK (LECTURE)
Cerebellum <i>more than motor</i> (HINDBRAIN)	Two hemispheres , each with three lobes (anterior, posterior, flocculonodular); midline vermis ; cerebellar cortex; four deep nuclei : fastigial, globose, emboliform, dentate (“Fat Guys Eat Donuts”)	Balance and coordination, plus many non-motor functions on the lecture’s function map: attention, working memory, verbal fluency, language, reasoning, autobiographical recall, saccades, hand movements	Medulloblastoma (children): walking, balance, motor, cognitive problems; 4th-ventricle block → hydrocephalus. Ipsilateral ataxia STEP 1

Cranial nerve nuclei by level **STEP 1**

Midbrain: III, IV · Pons: V, VI, VII, VIII · Medulla: IX, X, XI, XII. Neurotransmitter sources: **NE** = **locus coeruleus** (pons); **dopamine** = **SN + VTA** (midbrain); **serotonin** = **raphe**.

★ **USMLE high-yield**: Midbrain **tectum**: inferior colliculus = hearing, superior = vision; **tegmentum**: substantia nigra (Parkinson), VTA (reward), red nucleus (gait), PAG (pain), raphe (temperature). **Pons** = CN V–VIII nuclei + **locus coeruleus** (norepinephrine). **Medulla** = CN IX–XI nuclei + arcuate nucleus (breathing rate); runs breathing, heart rate, BP — **no survival without it** **EXAM EMPHASIS**. **Cerebellum** = 3 lobes per hemisphere, vermis, deep nuclei **Fat Guys Eat Donuts** (fastigial, globose, emboliform, dentate); many functions beyond motor.